

Euclid, Book I, Proposition I.1

To Construct an Equilateral Triangle on a Given Finite Straight Line

CGJteamLab

August 14, 2026

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1 The Classical Configuration and Construction

Let $A \neq B$. Euclid constructs two circles of radius AB : the first centered at A , the second centered at B . One of their points of intersection is denoted by C .

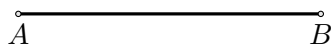
From the construction we obtain

$$AC \cong AB, \quad BC \cong AB.$$

Formally, however, this is not yet a complete proof. One must also show that the points A, B, C are not collinear.

1.0.1 Step 1. Given Segment

The construction starts with a given segment AB .



1.0.2 Step 2. Draw the Circle Centered at A

Draw the circle centered at A through B .

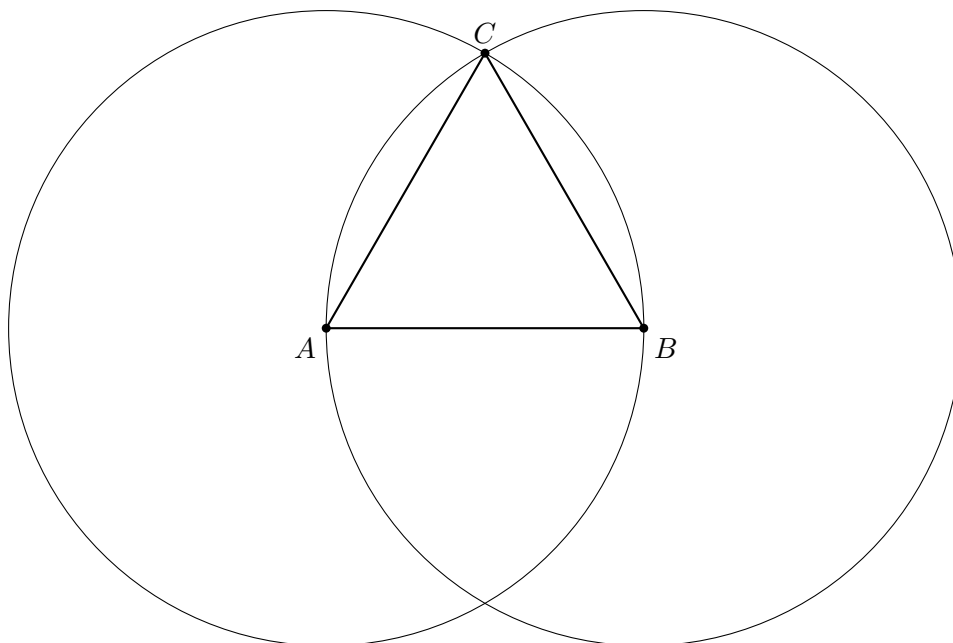
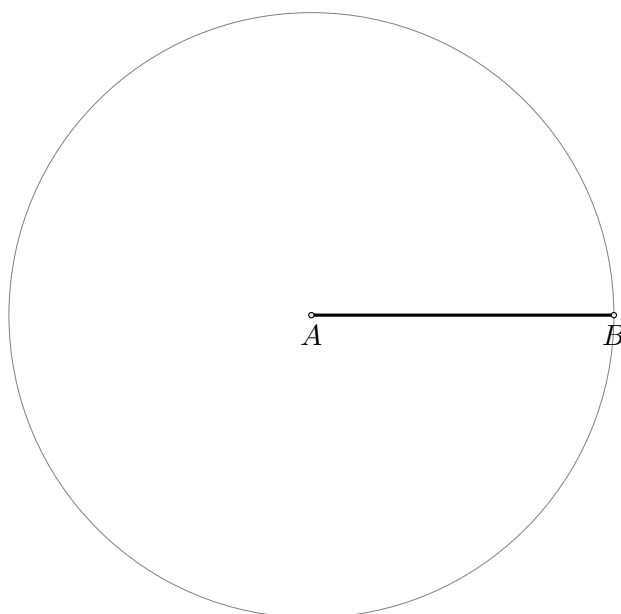
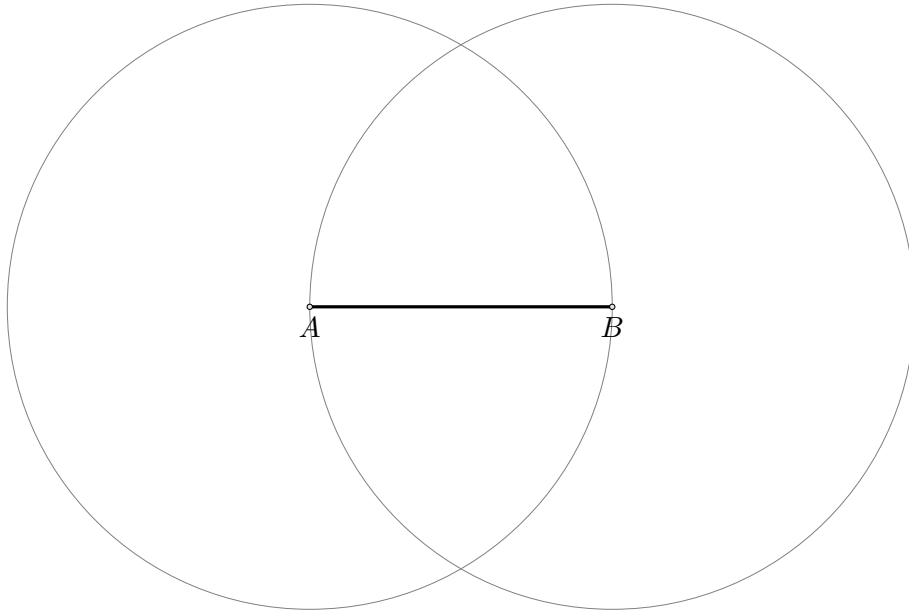


Figure 1: Euclid's classical construction. The point C is a common point of two circles of radius AB .



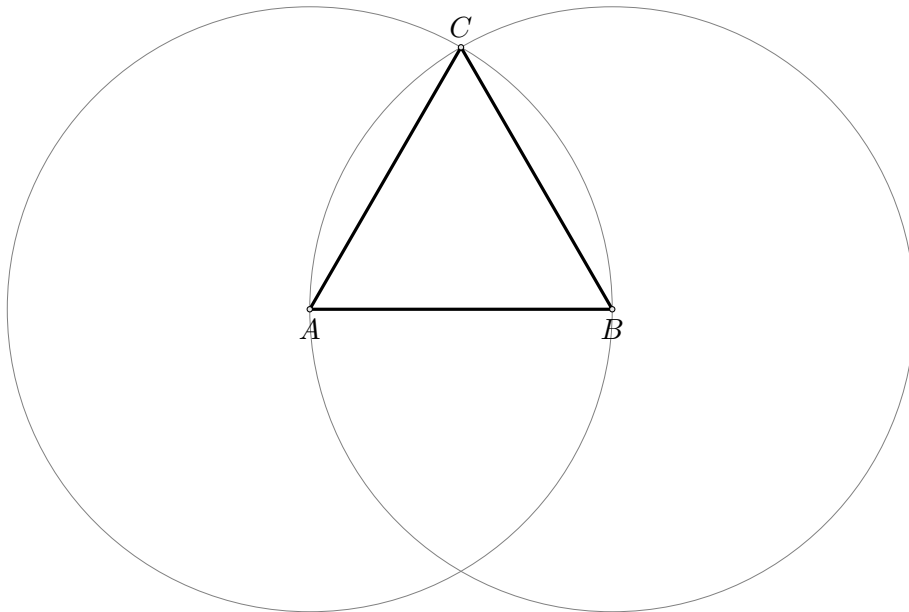
1.0.3 Step 3. Draw the Circle Centered at B

Draw the circle centered at B through A .



1.0.4 Step 4. Construct the Equilateral Triangle

Let C be one of the intersection points of the two circles. Join AC and BC to obtain the equilateral triangle.



2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \quad BC \cong AB.$$

In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg \text{Collinear}(A, B, C).$$

This stage no longer requires any theory of circles.

3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \quad BC \cong AB.$$

Then the points A, B, C are not collinear.

4 Proof

First observe that $C \neq A$ and $C \neq B$. If $C = A$, then $AC \cong AB$ would imply that a null segment is congruent to the non-null segment AB . Similarly, $C \neq B$.

Now suppose, for contradiction, that the points A, B, C are collinear. Since they are pairwise distinct, the trichotomy of the order of points on a line gives exactly three possibilities:

$$A - B - C, \quad B - A - C, \quad A - C - B.$$

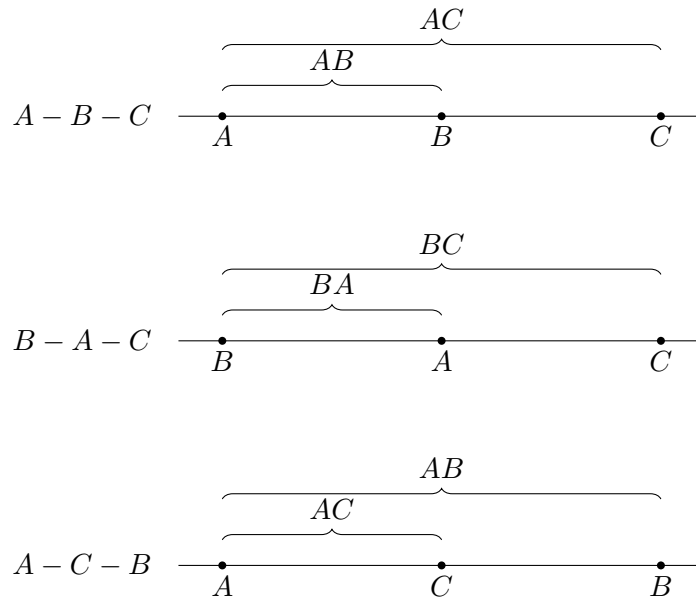


Figure 2: The three possible arrangements of pairwise distinct collinear points.

Case 1: $A - B - C$

The segment AB is a proper part of the segment AC . Book Zero contains the general theorem:

A proper part of a segment cannot be congruent to the whole segment.

On the other hand, by assumption,

$$AC \cong AB.$$

Using symmetry of congruence, we obtain

$$AB \cong AC,$$

which is a contradiction.

Case 2: $B - A - C$

The segment BA is a proper part of the segment BC . Since

$$BC \cong AB,$$

and reversing the endpoints of a segment does not change its length, we have

$$BA \cong AB.$$

By transitivity of congruence, it follows that

$$BA \cong BC,$$

again contradicting the fact that a proper part cannot be congruent to the whole.

Case 3: $A - C - B$

The segment AC is a proper part of the segment AB . But the assumption directly gives

$$AC \cong AB.$$

This yields the third contradiction.

All possible collinear arrangements lead to a contradiction. Therefore,

$$\neg \text{Collinear}(A, B, C).$$

□

5 Proof Architecture of the Reconstruction

6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

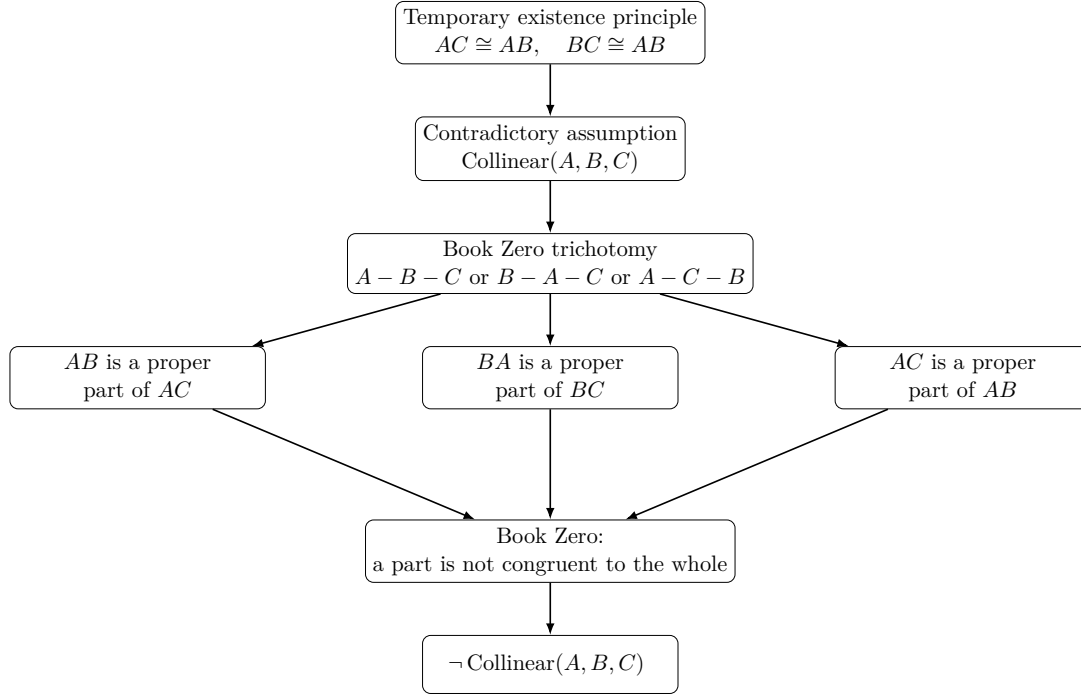


Figure 3: The structure of the proof of Proposition I.1 above the Book Zero layer.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

7 Relationship with the Foundations

Proposition I.1 has no earlier proposition of Book I on which it can depend. It therefore exposes the foundational boundary of the reconstruction more directly than any later proposition.

Euclid begins Book I with a circle-based construction: two circles with equal radii are drawn from the endpoints of the given segment, and an intersection point supplies the third vertex of the equilateral triangle.

In the Hilbert reconstruction, however, the essential issue is not segment congruence. Constructing points at prescribed distances is already supported by the congruence layer. The difficult step is to obtain a point realizing both distance conditions simultaneously and in the required nondegenerate configuration.

Thus I.1 immediately tests the interaction of

$$\text{incidence} \quad + \quad \text{order} \quad + \quad \text{congruence}$$

at the very beginning of Book I.

8 Hilbert and Book Zero Components Used

8.1 Segment Congruence

The equilateral conclusion is expressed synthetically:

$$AB \cong AC \quad \text{and} \quad AB \cong BC.$$

No numerical length function is used.

8.2 Betweenness Case Analysis

The reconstruction separates the possible collinear orders of three points. The three cases

$$A - B - C, \quad B - A - C, \quad A - C - B$$

are handled explicitly.

This is one of the earliest examples of a recurring formalization phenomenon: a configuration that is visually excluded in Euclid's diagram must be ruled out logically from order and congruence.

8.3 Book Zero Order Lemmas

The contradiction in each collinear branch depends on elementary results expressing that a proper part of a segment cannot be congruent to the whole segment.

These low-level theorems are precisely the kind of facts collected in Book Zero so that Book I proofs do not repeatedly reconstruct elementary betweenness arithmetic.

8.4 Noncollinearity

An equilateral triangle must be a genuine triangle. The formal construction therefore requires not only three pairwise congruent side relations but also a proof that the third point does not lie on the given line.

This nondegeneracy obligation is explicit in Lean and becomes an important pattern throughout the rest of Book I.

9 Formalization Notes

Proposition I.1 is unusually revealing because there is no preceding Euclidean theorem available to hide foundational work.

The classical diagram suggests immediately that the two circles meet away from the base line. A formal reconstruction must distinguish two questions:

Does a point satisfying the distance conditions exist?

and

Why is such a point noncollinear with the base endpoints?

The existing reconstruction isolates the second question and proves it synthetically. Assuming collinearity, the Hilbert order axioms reduce the configuration to the three possible betweenness orders. In each case, the equal-distance hypotheses force a proper part of a segment to be congruent to the whole, contradicting the established Book Zero segment-order theory.

Schematically,

assume collinearity $\longrightarrow$ three betweenness cases $\longrightarrow$ part/whole contradiction $\longrightarrow$ noncollinearity.

This is a foundationally significant result. It shows immediately why the Book Zero layer is useful: even the first proposition of the *Elements* depends, in a fully explicit proof, on elementary order facts that Euclid leaves to the diagram and common geometric understanding.

10 Dependency Summary

At the Euclidean level the construction is

given segment AB
 $\downarrow$
 circle centered at A with radius AB
 circle centered at B with radius BA
 $\downarrow$
 C at an intersection
 $\downarrow$
 $AC \cong AB, \quad BC \cong AB$
 $\downarrow$
 $\triangle ABC$ equilateral.

The formal reconstruction exposes an additional logical layer:

distance conditions for C
 $\downarrow$
 suppose A, B, C are collinear
 $\downarrow$
 $A - B - C \vee B - A - C \vee A - C - B$
 $\downarrow$
 Book Zero part-whole contradictions
 $\downarrow$
 $\neg \text{Collinear}(A, B, C).$

This is the first point in Book I where the formal dependency graph already extends below Euclid’s visible proposition sequence.

11 Conclusion

Proposition I.1 is the natural starting point of Euclid’s Book I, but its formal reconstruction immediately reveals that “starting point” does not mean “foundationally primitive.”

Euclid’s two-circle diagram suppresses elementary facts about point order, segment comparison, and nondegeneracy. Once these are made explicit, the proof reaches below Book I into the Hilbert and Book Zero layers.

The reconstruction therefore establishes an important theme for the entire project at the very first proposition: Book I is the visible classical development, while Book Zero supplies the elementary synthetic language needed to make its diagrammatic reasoning precise.

From this point onward, later propositions can reuse that language instead of rebuilding it. Proposition I.1 is consequently both the first Euclidean construction and the first demonstration of why the formal reconstruction requires a substantial interface beneath Euclid’s text.